



National
Qualifications
2025

2025 Biology

Higher - Paper 1

Question Paper Finalised Marking Instructions

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Marking instructions for each question

Question	Response	Mark
1.	D	1
2.	B	1
3.	C	1
4.	A	1
5.	B	1
6.	C	1
7.	C	1
8.	A	1
9.	B	1
10.	A	1
11.	D	1
12.	A	1
13.	D	1
14.	A	1
15.	D	1
16.	B	1
17.	D	1
18.	B	1
19.	B	1
20.	B	1
21.	C	1
22.	C	1
23.	D	1
24.	B	1
25.	C	1

[END OF MARKING INSTRUCTIONS]



National
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2025 Biology

Higher - Paper 2

Question Paper Finalised Marking Instructions

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Marking Instructions for each question

Question			Expected response	Max mark	Additional guidance
1.	(a)		Deoxyribose (sugar)	1	
	(b)	(i)	Any value or any range from 70 to 80 inclusive.	1	
		(ii)	Adds nucleotides to the primer/new strand. OR Replicates DNA/ target sequence.	1	
	(c)	(i)	DNA is replicated continuously. OR Nucleotides are added continuously. OR There are no fragments (to join). OR There is no lagging strand.	1	
		(ii)	It is not heat tolerant/resistant. OR It would be denatured.	1	
	(d)		1.6	1	
	(e)		17 920	1	

Question			Expected response	Max mark	Additional guidance
2.	(a)	(i)	Deletion	1	
		(ii)	All the amino acids after the mutation are changed/ affected. OR All the codons after the mutation are changed/affected which changes/affects the amino acids.	1	
	(b)		No/ineffective myostatin so no/ less inhibition of growth of muscle.	1	Must be clear that increased muscle growth is due to less inhibition by myostatin.
	(c)		Beef cattle have a longer length of pregnancy than dairy cattle. OR Dairy cattle have a shorter length of pregnancy than beef cattle. (1) Beef cattle have a lower birth mass than dairy cattle. OR Dairy cattle have a higher birth mass than beef cattle. (1)	2	
3.	(a)	(i)	Behavioural/reproductive	1	
		(ii)	Prevents gene flow/interbreeding. OR Prevents breeding between populations/groups.	1	
	(b)	(i)	Any value between 12000 and 14000 inclusive.	1	
		(ii)	The same fish may have been counted more than once. OR Not every fish passes the video cameras.	1	
		(iii)	Blue fish decrease and red fish increase.	1	

Question			Expected response	Max mark	Additional guidance
	(c)		They would not be able to interbreed/breed together to produce fertile offspring. OR Many/significant differences in base sequences.	1	

Question			Expected response	Max mark	Additional guidance
4.	(a)	(i)	Sequence data	1	
		(ii)	380	1	
		(iii)	2	1	
		(iv)	Last common ancestor of monocots and gymnosperms was more recent than monocots and lycopodiophyta. OR Monocots diverged more recently from gymnosperms than from lycopodiophyta.	1	
	(b)	(i)	Entire hereditary information encoded in DNA.	1	
		(ii)	Alternative (RNA) splicing occurs. OR Different combinations of exons are retained. (1) Different/multiple mature transcripts are produced. (1)	2	

Question			Expected response	Max mark	Additional guidance
5.	A		- transcription occurs in the nucleus - RNA polymerase unwinds the DNA/double helix - Hydrogen bonds break between the bases/strands - Complementary base pairing occurs - A primary transcript is formed OR The primary transcript contains introns and exons - Introns are removed/exons joined/retained in (RNA) splicing (Splicing)forms the mature transcript Any 4	4	Award marks for correctly labelled diagrams.
	B		- Duplication adds gene(s)/section of chromosome from/to its homologous partner/chromosome - Deletion removes gene(s)/section of chromosome - Inversion is where a gene(s)/section of chromosome is reversed/flipped/rotated 180° - Translocation adds gene(s)/section of chromosome from a non-homologous/different chromosome - Duplication is important in evolution OR Duplication allows beneficial mutations to occur in duplicated gene while original gene is still expressed - chromosome mutations can be lethal - Naming all 4 mutations - Only award if none of points 1-4 are awarded. Any 4	4	

Question			Expected response	Max mark	Additional guidance
6.	(a)	(i)	Catabolic	1	
		(ii)	Description: Active site changes shape. (1) Explanation: Lowers the activation energy. OR Lowers the energy required for the reaction. (1)	2	
	(b)		Description: Binds to/inhibits enzyme 1/earlier enzyme. (1) Advantage: Saves energy/resources. OR Prevents over-production/build-up of end product/isoleucine. (1)	2	
	(c)	(i)	All active sites are occupied. OR No more active sites available.	1	
		(ii)	Inhibition is not reversed/reduced/overcome by increasing/high substrate concentration. OR Increasing substrate concentration does not affect the rate of reaction at high substrate concentrations. OR Same maximum rate of reaction is not reached by increasing substrate concentration.	1	

Question			Expected response	Max mark	Additional guidance
7.	(a)	(i)	Inner mitochondrial membrane	1	
		(ii)	ATP synthase	1	
		(iii)	Y oxygen/O ₂ (1) Z water/H ₂ O (1)	2	
	(b)		Carries/transfers/hydrogen ions and electrons. (1) From glycolysis/citric acid cycle/cytoplasm/matrix of mitochondria. OR To electron transport chain/inner mitochondrial membrane. (1)	2	
	(c)		Fewer electrons move along the electron transport chain/carrier proteins releasing energy. OR Less energy to move hydrogen ions across the membrane. OR Less hydrogen ions pumped/actively transported across the membrane. (1) Less hydrogen ions to pass through/drive/rotate ATP synthase. (1)	2	
8.	(a)	(i)	From 2.0 to 2.4 m/s it stays constant at 1.4 mmol/L. (1) From 2.4 m/s it increases to 6.3 mmol/L. (1)	2	Award 1 mark for remains constant to 2.4m/s then increases.
		(ii)	1.7	1	
		(iii)	47.5/47.54/47.541	1	
	(b)	(i)	0.8	1	
		(ii)	4.4	1	
	(c)		Fermentation	1	
	(d)		Lactate is converted to pyruvate.	1	

Question			Expected response	Max mark	Additional guidance
9.	(a)	(i)	10 : 3 : 8	1	
		(ii)	Plants grown in soil with arsenic + ARB have lower pigment/chlorophyll/ carotenoid levels than plants grown in soil with no arsenic.	1	
		(iii)	Increase/extend the range/number of wavelengths/ colours of light absorbed. OR Broadens absorption spectrum. OR Absorbs wavelengths/light/colours of light not absorbed by chlorophyll. (1) Energy passed on to chlorophyll. (1)	2	
	(b)		To excite electrons (in pigment molecule). (1) Transfer of the electrons along/through the electron transport chain releases energy. (1) ATP is generated by ATP synthase. (1) Any 2 from 3	2	

Question			Expected response	Max mark	Additional guidance
10.	(a)	(i)	They compete for light/space/minerals/water/ nutrients.	1	
		(ii)	Weeds have broader leaves so absorb more chemical.	1	
	(b)	(i)	60	1	
		(ii)	More people/population increasing. (1) More resistant weeds so less crops/ yield. OR More resistant weed/species so more competition with crops/wheat. (1)	2	
	(c)		Resistant (weeds) survive or non-resistant (weeds) die. (1) pass on resistance alleles/genes to offspring/next generation. (1)	2	

Question			Expected response	Max mark	Additional guidance
11.	(a)	(i)	Sumithion concentration	1	
		(ii)	Temperature of water/solution. OR pH/salinity of water/solution. OR Species/type of eggs/fish.	1	
		(iii)	0.0/0 (1) Shows that sumithion was reducing hatching success. OR Shows the effect of sumithion (on hatching success). OR To compare with sumithion. (1)	2	
	(b)		Axes correctly labelled and scale correct. (1) Points correctly plotted and joined. (1)	2	
	(c)		As sumithion concentration increased hatching success decreased (until 0.8 mg/L) then remained the same.	1	
	(d)		Bioaccumulation/biomagnification. OR Persistence in the environment. OR Results in resistant pest (populations).	1	

Question			Expected response	Max mark	Additional guidance
12.	(a)	(i)	210	1	
		(ii)	There are range bars	1	
		(b)	F ₂ show variation	1	
	(c)	(i)	Cost effective. OR Cheap. OR Less land/labour. OR High profits. OR High productivity.	1	
		(ii)	There is increased/high chance of transmission by direct contact. OR Overcrowding so parasites are spread by direct contact.	1	
		(iii)	Hysteria	1	
13.	(a)		Fly offspring benefit/gain while the bee larvae are harmed/lose. (1) by gain/loss of pollen/nutrients/energy/food. (1)	2	
	(b)		Worker (bees) defend the hive/larvae.	1	
	(c)		Ensures shared genes survive/will be passed on. OR Increases survival of shared genes.	1	

Question			Expected response	Max mark	Additional guidance
14.	(a)	(i)	Co-operative hunting	1	
		(ii)	Can catch/kill larger prey. OR Less energy used per individual.	1	NOT hunt larger prey.
	(b)		7.1	1	
	(c)		Method: Habitat corridors. (1) Explanation: Increased access to food/mates. OR Recolonisation (after local extinction). (1)	2	

Question			Expected response	Max mark	Additional guidance
15.	A		1. animals survive adverse conditions by using dormancy 2. dormancy/hibernation/aestivation/daily torpor reduces metabolic rate 3. dormancy/hibernation/aestivation/daily torpor reduces heart rate/breathing rate/body temperature 4. dormancy/hibernation/aestivation/daily torpor saves energy 5. dormancy/hibernation/aestivation/daily torpor allows survival when costs of normal metabolic activity would be too high 6. predictive dormancy before (onset of) adverse conditions 7. consequential dormancy after onset of adverse conditions 8. hibernation in winter/low temperature AND aestivation in high temperature/drought 9. daily torpor in animals with high metabolic rates Any 7	7	NOT avoid adverse conditions using dormancy.

Question			Expected response	Max mark	Additional guidance
	B		1. all 4 phases named: lag, log/exponential, stationary and death 2. lag phase when enzymes are induced OR lag phase when DNA replicates 3. log/exponential phase when fast/most growth occurs 4. log/exponential phase when nutrients are plentiful 5. stationary phase when nutrients are being depleted/used up OR toxic metabolites/toxins produced/build up in stationary phase 6. secondary metabolites/antibiotics are produced/build up in stationary phase 7. antibiotics/secondary metabolites confer ecological advantage OR antibiotics/secondary metabolites reduce/remove competition from other microorganisms 8. death phase occurs due to toxic accumulation of metabolites/nutrients have run out 9. viable cell count is only living micro-organisms OR total cell count is living and dead micro-organisms Any 7	7	

[END OF MARKING INSTRUCTIONS]

General marking principles for Higher Biology

Always apply these general principles. Use them in conjunction with the marking instructions, which identify the key features required in candidates' responses.

- (a) Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.
- (b) If a candidate response does not seem to be covered by either the principles or detailed marking instructions, and you are uncertain how to assess it, you must seek guidance from your team leader.
- (c) Do not award half marks.
- (d) Where a candidate makes an error in the first part of a question, award marks for subsequent answers that are correct with regard to this original error. Do not penalise candidates more than once for the same error.
- (e) Unless a numerical question specifically requires evidence of working to be shown, award full marks for a correct final answer (including units, if appropriate) on its own.
- (f) Candidates should not use bulleted lists to answer extended-response questions. They must respond to the 'command' word as appropriate and provide extended answers to communicate fully their knowledge and understanding. Candidate responses in the form of bulleted lists may not be able to access the full range of available marks.
- (g) In the detailed marking instructions, if a word is underlined then it is essential; if a word is **(bracketed)** then it is not essential.
- (h) In the detailed marking instructions, words separated by / are **alternatives**.
- (i) A correct response can be negated if the candidate includes:
 - an extra, incorrect, response
 - additional information that contradicts the correct response
- (j) Where the candidate is instructed to choose one question to answer but instead answers two questions, mark both responses and award the higher mark.
- (k) Unless otherwise required by the question, the use of abbreviations (for example DNA, ATP) or chemical formulae (for example CO₂, H₂O) are acceptable alternatives to naming.
- (l) If a numerical answer is required and units are not given in the stem of the question or in the answer space, candidates must supply the units to gain the mark. If units are required on more than one occasion, do not penalise candidates repeatedly.
- (m) If incorrect spelling is given:
 - If the correct word is recognisable then award the mark.
 - If the word can easily be confused with another biological term then **do not** award the mark, for example glucagon and glycogen.

(n) Presentation of data:

- If a candidate provides two graphs, in response to one question, mark both and award the higher mark.
- If a question asks for a particular type of graph/chart and the candidate gives the wrong type, do not award full marks. Candidates cannot achieve the plot mark but **may** be able to achieve the mark for scale and label. If the x and y data are transposed, then do not award the scale and label mark.
- If the graph uses less than 50% of the axes then do not award the scale and label mark.
- If 0 is plotted when no data for this is given, then do not award the plot mark – candidates should only plot the data given.

(o) Only award marks for a valid response to the question asked. For example, in response to questions that ask candidates to:

- **identify, name, give or state**, they need only answer or present in brief form
- **describe**, they must provide a statement as opposed to simply one word
- **explain, justify**, they must provide a reason for the information given
- **compare**, they must demonstrate knowledge and understanding of the similarities and/or differences between topics being examined
- **calculate**, they must determine a number from given facts, figures or information
- **predict**, they must indicate what may happen based on available information
- **suggest**, they must apply their knowledge and understanding to a new situation

2025 Higher Biology Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	D	11	10	6	73	0	0
2	B	2	39	52	6	0	0
3	C	29	20	33	19	0	0
4	A	52	23	8	17	0	0
5	B	3	70	5	21	0	0
6	C	6	3	89	2	0	0
7	C	8	7	80	5	0	0
8	A	49	10	24	18	0	0
9	B	10	81	2	7	0	0
10	A	84	4	6	6	0	0
11	D	4	10	14	71	0	0
12	A	90	4	4	2	0	0
13	D	5	6	12	76	0	0
14	A	93	4	1	3	0	0
15	D	15	20	7	58	0	0
16	B	6	74	12	7	0	0
17	D	10	17	9	64	0	0
18	B	10	76	8	5	0	0
19	B	3	77	16	4	0	0
20	B	20	31	27	22	0	0
21	C	5	22	69	4	0	0
22	C	9	7	82	2	0	0
23	D	9	2	11	78	0	0
24	B	14	60	3	23	0	0
25	C	9	7	81	3	0	0

2025 Higher Biology Section 2 Marks Data Analysis

2025 Item	Max Mark	Average Mark	% Average Mark	Discrimination	% Not Attempted
Q1a	1	0.6	62.1%	39	1.4%
Q1bi	1	0.8	81.4%	37	0.2%
Q1bii	1	0.6	58.1%	40	1.6%
Q1ci	1	0.5	51.8%	55	4.6%
Q1cii	1	0.7	67.1%	49	7.4%
Q1d	1	0.7	71.1%	41	2.0%
Q1e	1	0.7	72.8%	41	1.1%
Q2ai	1	0.8	80.0%	27	0.6%
Q2aii	1	0.3	32.9%	48	1.7%
Q2b	1	0.3	33.0%	47	5.0%
Q2c	2	1.7	84.6%	14	0.6%
Q3ai	1	0.5	50.0%	34	2.0%
Q3aii	1	0.4	39.6%	51	2.9%
Q3bi	1	0.7	71.9%	19	2.4%
Q3bii	1	0.9	85.4%	20	1.2%
Q3biii	1	0.9	85.1%	11	0.5%
Q3c	1	0.5	54.9%	47	1.6%
Q4ai	1	0.6	58.1%	53	4.6%
Q4aii	1	0.8	80.0%	34	0.2%
Q4aiii	1	0.6	64.8%	19	0.4%
Q4aiv	1	0.8	77.1%	38	0.8%
Q4bi	1	0.6	58.0%	49	4.6%
Q4bii	2	0.7	34.0%	59	11.9%
Q5A	4	2.4	59.4%	67	10.1%
Q5B	4	2.1	52.9%	69	0.0%
Q6ai	1	0.8	76.0%	38	3.4%
Q6aii	2	0.8	37.6%	49	6.1%
Q6b	2	0.8	37.6%	63	11.7%
Q6ci	1	0.1	14.9%	32	6.3%
Q6cii	1	0.3	28.4%	44	5.8%
Q7ai	1	0.7	69.2%	56	1.4%
Q7aii	1	0.7	71.3%	59	2.5%
Q7aiii	2	1.2	60.0%	56	2.8%
Q7b	2	0.9	44.4%	59	7.0%
Q7c	2	0.3	15.4%	50	11.4%
Q8ai	2	1	49.4%	53	1.3%
Q8aii	1	0.6	55.6%	49	0.5%
Q8aiii	1	0.5	54.0%	47	2.4%
Q8bi	1	0.2	22.2%	46	1.9%
Q8bii	1	0.5	53.8%	49	1.7%
Q8c	1	0.7	71.4%	45	3.0%
Q8d	1	0.2	15.5%	38	4.3%
Q9ai	1	0.5	45.2%	42	2.5%
Q9aii	1	0.4	38.6%	38	7.5%
Q9aiii	2	0.6	31.7%	60	6.5%
Q9b	2	0.8	42.1%	64	11.1%
Q10ai	1	0.6	58.4%	16	1.9%
Q10aii	1	0.3	28.7%	46	6.1%
Q10bi	1	0.9	89.8%	27	1.2%
Q10bii	2	1.1	55.5%	37	1.5%
Q10c	2	0.9	43.2%	50	4.1%
Q11ai	1	0.8	81.9%	37	0.8%
Q11aii	1	0.5	51.9%	14	1.4%
Q11aiii	2	1.5	77.2%	47	1.9%
Q11b	2	1.7	84.9%	33	1.1%
Q11c	1	0.2	21.3%	15	1.0%
Q11d	1	0.6	60.7%	52	3.1%
Q12ai	1	0.4	36.8%	34	4.3%
Q12aii	1	0.7	70.8%	15	3.9%
Q12b	1	0.4	40.0%	51	5.9%
Q12ci	1	0.9	91.5%	32	1.3%
Q12cii	1	0.2	21.2%	41	1.6%
Q12ciii	1	0.5	54.6%	53	7.7%
Q13a	2	1.3	63.2%	49	3.3%
Q13b	1	0.4	37.3%	43	3.4%
Q13c	1	0.3	26.5%	55	7.1%
Q14ai	1	0.7	69.0%	46	3.4%
Q14aii	1	0.6	58.4%	47	0.8%
Q14b	1	0.4	35.2%	31	1.3%
Q14c	2	1.2	60.0%	60	8.7%
Q15A	7	4.2	59.8%	74	2.0%
Q15B	7	5.1	72.6%	68	0.0%
Paper	95	51.8	54.5%		

2025 Higher Biology Section 2 Frequency of Marks Analysis

2025 Item	0	1	2	3	4	5	6	7
Q1a	38	62						
Q1bi	19	81						
Q1bii	42	58						
Q1ci	48	52						
Q1cii	33	67						
Q1d	29	71						
Q1e	27	73						
Q2ai	20	80						
Q2aai	67	33						
Q2b	67	33						
Q2c	12	8	81					
Q3ai	50	50						
Q3aai	60	40						
Q3bi	28	72						
Q3bii	15	85						
Q3biii	15	85						
Q3c	45	55						
Q4ai	42	58						
Q4aai	20	80						
Q4aiii	35	65						
Q4aiv	23	77						
Q4bi	42	58						
Q4bii	49	34	17					
Q5A	21	12	14	14	39			
Q5B	23	18	14	13	32			
Q6ai	24	76						
Q6aai	37	51	12					
Q6b	48	29	23					
Q6ci	85	15						
Q6cii	72	28						
Q7ai	31	69						
Q7aai	29	71						
Q7aiii	36	9	56					
Q7b	42	27	31					
Q7c	78	14	8					
Q8ai	41	19	40					
Q8aai	44	56						
Q8aiii	46	54						
Q8bi	78	22						
Q8bii	46	54						
Q8c	29	71						
Q8d	84	16						
Q9ai	55	45						
Q9aai	61	39						
Q9aiii	51	34	15					
Q9b	45	25	29					
Q10ai	42	58						
Q10aai	71	29						
Q10bi	10	90						
Q10bii	12	66	23					
Q10c	40	34	26					
Q11ai	18	82						
Q11aai	48	52						
Q11aiii	10	26	64					
Q11b	6	18	76					
Q11c	79	21						
Q11d	39	61						
Q12ai	63	37						
Q12aai	29	71						
Q12b	60	40						
Q12ci	9	91						
Q12cii	79	21						
Q12ciii	45	55						
Q13a	21	32	47					
Q13b	63	37						
Q13c	74	26						
Q14ai	31	69						
Q14aai	42	58						
Q14b	65	35						
Q14c	20	41	40					
Q15A	11	7	9	10	12	13	14	24
Q15B	1	5	6	10	13	14	13	37

2025 Higher Biology Section 3 Assignment Marks Analysis

Item	Max Mark	Average Mark	% Average Mark	Discrimination
1	1	0.8	84.9%	28
2	4	2.5	63.0%	44
3a	1	0.5	54.3%	30
3b	1	1	95.4%	3
3c	1	0.5	53.3%	33
3d	1	0.7	65.8%	25
3e	1	0.5	50.1%	33
4a	1	1	97.8%	16
4b	1	0.9	85.3%	24
4c	1	0.9	90.2%	21
4d	1	0.8	76.9%	28
5	1	0.3	33.0%	41
6	1	0.5	48.2%	30
7	3	1.4	46.0%	50
8	1	0.8	82.8%	19
Paper	20	13.1	65.4%	

Item	Max Mark	0	1	2	3	4
1	1	15	85			
2	4	10	15	21	21	33
3a	1	46	54			
3b	1	5	95			
3c	1	47	53			
3d	1	34	66			
3e	1	50	50			
4a	1	2	98			
4b	1	15	85			
4c	1	10	90			
4d	1	23	77			
5	1	67	33			
6	1	52	48			
7	3	24	31	28	17	
8	1	17	83			
Paper	20					

Item	Max Mark	Item Description
1	1	An aim that describes clearly the purpose of the investigation
2	4	Underlying biology
3a	1	A brief summary of the approach used to collect experimental data.
3b	1	Sufficient raw data from the candidate's experiment.
3c	1	Data, including mean/average values, presented in a correctly produced table(s)
3d	1	Data relevant to the aim from an internet/literature source
3e	1	A citation and reference for a source of the internet/literature data
4a	1	An appropriate format from the options of bar graph or line graph.
4b	1	The axes of the graph have suitable scales.
4c	1	The axes of the graph have suitable labels and units
4d	1	Accurately plotted data points and a line (line graph) or clear bar tops (bar graph)
5	1	A valid comparison or an appropriate calculation linked to the aim of the investigation.
6	1	A valid conclusion that relates to the aim and is supported by all the data in the report.
7	3	Evaluation of the investigation
8	1	A clear and concise report, with an informative title.